

THE IMPACT OF CONTACT

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Abstract—Scenarios of the impact on human society of radio contact with an extraterrestrial civilization vary widely between paranoid projections that contact with advanced extraterrestrials would quickly devastate the human spirit, and pronoid predictions that the extraterrestrials would be so advanced that they would swiftly and benevolently lead us into a golden age. Such extreme pronouncements exaggerate the probable speed and magnitude of the impact of radio contact in that they conflate technical and cultural understanding, and ignore the problem of inter-civilizational comprehension. Judging from the record of cultural misunderstanding between closely related human groups, comprehending a totally different civilization light years away, and absorbing the meaning of whatever messages were sent, would be a slow and tedious process calling for the efforts of specialists from many disciplines as well as the SETI scientists now engaged in the search.

1. INTRODUCTION

Forecasting how contact with an extraterrestrial civilization will impact human society is basically an impossible task. We do not know the nature of the civilization, or civilizations, with which we might come into contact. Indeed, their very existence is only hypothesized. Nor, probably, do we really know enough about ourselves to say how we would react to contact even if we did know the nature of our potential galactic partner or partners. Ignorance, however, has hardly stopped speculation about the impact of contact. If anything, it has promoted a diversity of opinion of what will happen to human society if and when contact is made.

Perhaps the most common element in these forecasts is the premise that any civilization with which we are likely to come into contact in the near future would probably be more advanced than we. That follows from the argument that should intelligent life abound the galaxy, we are hardly likely to be the oldest and most advanced civilization. Given our only recently developed, still rudimentary scientific and technological capabilities, the consensus is that we are likely to be one of the least advanced civilizations in the galaxy. Indeed, this assumption is built into our essentially passive efforts to establish contact - either through trying to detect the appearance of alien spaceships that might be reconnoitering Earth, or through building ever more sensitive receivers and signal analyzers to detect any messages that might be being beamed toward us.

Given the impact of colonial powers on technologically less advanced populations around the world, and how we are now driving toward extinction the intellectually most able of our relatives, the chimpanzee and the gorilla, it is not surprising that paranoid visions of the impact on human society of contact with

advanced extraterrestrials abound. According to the scenarios, we are doomed, or at least will suffer massive cultural and psychological disruption, should contact with a more advanced galactic neighbor occur.

Not all forecasters are so paranoid, however. Some are so optimistic as to merit the label "pronoid," a recently coined antonym for paranoid which is used to refer to individuals who assume that everyone around them has their welfare at heart. Such theorists propose that the extraterrestrials who will contact us would be behaviorally superior to us, as well as technologically more advanced. They imagine that these super-intelligent and hyper-benevolent beings will gladly tell us how to solve the secrets of nature, and patiently show us how we may survive our nuclear adolescence and learn to live together in peace.

While for the purposes of this symposium (in which it is assumed that contact will be made via radio transmissions and not face-to-face encounters) much of this speculation may be dismissed as so much science-fiction of the *War of the Worlds* genre, or as merely the wishful thinking of UFO enthusiasts, there are nonetheless some equally paranoid, and pronoid, speculations of what might follow radio contact with an extraterrestrial civilization.

The realization via radio signals that we are indeed at a rudimentary stage of development, and the intellectual dependency upon the superior civilization which has contacted us that would follow, would be absolutely shattering according to a number of writers. The psychologist Jung, for example, declared that "the reins would be torn from our hands and we would, as a tearful old medicine man once said to me, find ourselves 'without dreams,' that is we would find our intellectual and spiritual aspirations so outmoded as to leave us completely paralyzed[1]." By no means do all

such paranoid pronouncements come from humanists. For example, at least three Nobel laureates, Joshua Lederberg[2], Martin Ryle[3] and George Wald[4] have expressed their misgivings about radio contact with extraterrestrials. Wald, for example, writes that "the thought that we might attach, as by an umbilical cord, to some more advanced civilization, with its more advanced science and technology, in outer space does not thrill me, but just the opposite. I can conceive of no nightmare," he declares, "as terrifying as establishing such communication with a so-called superior (or if you wish, advanced) technology in outer space."

In contrast, the pronoid school of thought says we have much to gain through establishing radio contact with an advanced civilization, and then receiving our fill of their more advanced knowledge and wisdom[1,5]. One of the most comprehensive statements of this was penned by the chemist and supporter of the Search for Extraterrestrial Intelligence (SETI) Harrison Brown in his novel of radio contact, *The Cassiopeia Affair*[6]. He puts the following words into the mouth of his fictional president of the United States following the reception of a signal from Cassiopeia 3579.

"Think of the wonderful things we can learn from our brothers of Cassiopeia. The chemistry of their life processes, the properties of their planet and their planetary system, the Galaxy as viewed from their location - all these possibilities bring tremendous excitement to our scientists.

"We believe that Cassiopeian civilization is technologically more advanced than our own. Their knowledge may enable us to leapfrog over centuries of development. Cassiopeia 3579 might well provide us a sort of cosmic technology assistance program for this emerging earth.

"Above all, our Cassiopeian brothers may have found a way to solve political problems comparable to those which now threaten us on earth. They may have found ways of managing such problems without precipitating a planet-wide holocaust such as we have feared on earth for so long."

As an anthropologist who has recently been studying the space community, I have been amazed at the sweep of imagination of those dedicated to space ventures. Now that we have learned to send manned rockets into space, say

some space settlement enthusiasts, exponential human expansion throughout the galaxy is both inevitable and close at hand[7]. SETI advocates can match this hubris when it comes to imagining how easy it will be to comprehend an extraterrestrial civilization once a signal has been received, and to reap whatever rewards that comprehension will bring. They assume that extraterrestrials are beaming a signal toward us which has been deliberately "anti-encrypted" for ease of decipherment. Some further posit that the signal will contain a tutorial based on universal mathematical truths and physical constants to instruct us how we may readily tap into a hitherto undreamed of wealth of knowledge.

2. CULTURAL MISUNDERSTANDING

I think that many SETI theorists vastly overestimate the ease of comprehending extraterrestrial transmissions, and their consequent impact, because they conflate technical and cultural understanding. My experience as an anthropologist studying the contact of formerly isolated terrestrial cultures leads me to be much more alert to the difficulties of inter-group communication than many SETI advocates seem to be. Misunderstandings abound on the frontier of culture contact. Let me cite an especially perplexing example.

New Guinea was first settled some 50,000 years ago when glaciation greatly lowered sea levels, enabling small groups of people from Southeast Asia to raft across the consequently narrowed ocean straits. With rising sea level in the Holocene, New Guinea became harder and harder to reach from the Asian mainland. Except for the infusion of small groups of canoe voyagers sailing from Indonesia and the Philippines, the New Guineans remained essentially isolated from the rest of the world until late in the era of European colonial expansion. Significant contact did not really come until early in this century when Australians, British, Dutch and Germans began to trade and extend political control along the coast.

The gulf between these stone age tribesmen and the intruding agents of Western colonials amounted to more than just a gap in technology. After initial contact, and a few armed encounters, trade in foodstuffs and other local products for axes and other steel tools was initiated, followed by the development of

foreign-owned plantations and the planting of cash crops by the New Guineans themselves. However, to the consternation of the traders and colonial officials, a number of groups which had started along this path of colonial development soon quit working on the plantations, stopped tending their own crops, and started behaving in ways considered most bizarre by the Western intruders[8].

Villagers living on the coast would build a crude wharf, and then erect a "radio station" - made of wood, bamboo and vines. They then would spend days dancing, carrying out elaborate rituals, gathering at the "radio station" to send messages, and waiting on the wharf for a ship to come. Other villagers living inland, where rickety biplanes represented the epitome of modern technology brought by the outsiders, would hack a crude airstrip out of the jungle, and then carry similarly mysterious rituals designed to attract an airplane to land on their strip. These cults which swept through New Guinea and adjacent islands came to be known as "cargo cults" in that the wharfs, radio stations, airstrips, and the associated rituals and activities were all meant to attract ships and planes laden with "cargo," that is, the tools, tinned food, clothing, and items of wealth and technological wizardry which Westerners had under their control.

Once these cults gripped the people, the colonial officials found it virtually impossible to talk them out of their beliefs. The officials would explain, for example, that metal tools, clothing, as well as more complicated examples of early twentieth century high technology, were made in factories and could only be purchased by money, not conjured up by ritual. Cargo cult adherents did not believe them, however. They had worked hard on the plantations, and had tried growing their own cash crops, yet with the small sums of money they earned from these efforts they found that they could buy very little in the way of modern goods. At the same time, they saw that Westerners could, through making little markings on thin white sheets of material, or through talking into a mysterious apparatus, call forth ships and planes which would bring them an abundance of goods. They felt that Westerners were lying to them, that they were making them do all the hard work while at the same time denying to them the true secret of obtaining "cargo"[9].

A massive cultural misunderstanding, born of the sudden contact of technologically and epistemologically alien peoples, was therefore at the root of these cults. This misunderstanding,

which took decades to clear up, was shared by both sides. The Westerners, imbued with their ideas of colonial progress, could not understand how anyone could not believe their exhortations that the road to the wealth and affluence of the modern world lay through hard work in the plantations and cash crop gardens. The New Guineans, in turn, were sure they were being cheated and fell back on their own cosmology to explain the deception and to take action to counteract it. In the beginning, they believed, it was the gods who created all technology and gave it to the people. So too, they reasoned, must the gods have created the new technology brought by the intruders who, somehow, had gained access to these latest developments from the gods and were denying the New Guineans that which should rightfully be theirs. Hence, the wharf building, the gathering at bush "radio stations," and other activities which Westerners found so bizarre, but which New Guineans believed was the only way in which they could gain direct access to "cargo."

3. COMPREHENDING AN EXTRATERRESTRIAL CULTURE

It is tempting to suggest that some SETI enthusiasts, particularly those who wax ecstatically about all the marvelous advantages to humanity that will flow from contact with extraterrestrials, may already be engaged in a kind of high technology cargo cult. If only, they say, we could build large enough antennas, sensitive enough receivers, and sophisticated enough code analyzers, we could intercept the messages beamed toward us and tap the rich knowledge of our more advanced galactic neighbors. However, I raise the example of the cargo cult not to make a facile comparison, but to point out the difficulties inherent in establishing meaningful communication between alien groups.

If populations of the same species who are in face-to-face contact can disagree over such a basic issue as the derivation of technology, what might happen when we come into radio contact with another intelligent species from a distant star system with a biological basis, evolutionary history, and cultural development entirely independent from our own? If slight variants within our own species have difficulty understanding one another, what will it be like trying to comprehend an entirely unrelated extraterrestrial civilization through the medium of one-way radio transmissions, or at best

tortuously slow two-way "conversations"? The discovery of patterned radio signals, and that an apparently shared knowledge of universal math and science principles enables us to begin to make some sense of them, would be just the beginning. The really hard task of comprehending a totally alien culture would remain.

Lest this be dismissed as the misguided thoughts of someone with no deep understanding of radiophysics or how to construct an algorithm for discriminating artificial from natural signals, I refer to the writings of one of the founding fathers of the SETI movement, Philip Morrison. While Morrison may posit that anti-encrypted messages are being beamed to us, and that shared mathematical and scientific knowledge may enable us to begin to decipher them, he does not think that understanding the nature of the senders and their civilization will come easy.

After the initial headlines over the reception of a message subside, the long and hard struggle to make sense of what is being transmitted will begin, says Morrison. "The recognition of the signal," he writes, "is the great event, but the interpretation of the signal will be a social task comparable to that of a very large discipline, or branch of learning. Most of this very complex signal will contain," he continues, "not mainly science and mathematics but mostly what we would call art and history[10]." Such cultural material would not be easy to decipher, much less to understand fully. In Morrison's view, the process would continue for generations, and would be the stuff of books and universities not newspapers and daily news broadcasts[11].

4. ASTROSOCIOLOGY AND THE ANTICIPATION OF CONTACT

Morrison's reasoning suggests that social scientists and others concerned with the human condition should join in the SETI effort. Currently, however, SETI is dominated by workers from only one of C. P. Snow's two cultures: astronomers, physicists, radio engineers, computer scientists and the like. This is probably how it should be at this early stage of the search. After all, unless the SETI scientists can develop the apparatus needed to receive and recognize an extraterrestrial signal, there will be nothing for anyone else to do.

But, what about the future? Should social scientists and humanists anticipate the reception of a signal, and the task of interpreting an

extraterrestrial culture, by laying the groundwork for the comparative study of cosmic civilizations, as a number of Russian scholars urge[12]? In other words, should they join together to form a new inter-disciplinary field, which might be called "xenosociology" or "astrosociology"?

Actually, I doubt if many social scientists and humanists would now be interested in developing such a field. Social scientists and humanists have so far expressed minimal interest in space exploration and development[13], and it seems unlikely that a new field dedicated to the analysis of hypothesized extraterrestrial civilizations which may or may not exist somewhere in space would be very attractive to them. Nonetheless, the case for considering the possibility of extraterrestrial civilizations, and for developing methods for studying and communicating with any such civilizations, needs to be pushed, for such an effort might pay off in several ways.

First, even in if we did not make contact, we might well benefit from an extraterrestrial perspective. The recent stimulus investigations about such extraterrestrial phenomena as cometary showers have had on thinking about the evolution of life on Earth provides a model[14]. Just thinking about how other intelligent civilizations might develop, and about how we might relate to such civilizations, could stimulate us to take a detached and non-ideological perspective on our own global civilization and its problems. It is often said that those first photos of earth taken from space made us fully realize the significance and value of our ecosphere. A view from an extraterrestrial civilization, even an imaginary one, might help us to transcend our cultural conceits and political divisions and think constructively about our own global civilization.

Second, if contact were to occur, we would be better prepared for the challenge of cultural interpretation. That challenge might even turn out to be more immediate than I have so far suggested. Why should an advanced civilization attempt to contact us to learn our rudimentary science and technology, or merely to export their more advanced version? Anthropological investigation, not technology transfer, might be a more likely motive. To paraphrase Morrison[15], the extraterrestrial's "Department of Anthropology" might be in charge, directing an effort aimed at learning the nature of our culture as well as informing us of theirs. If so, in addition to all the astronomers, radio engineers, computer scientists and others technically

qualified to receive and analyze signals, we would certainly need a corps of anthropologists, sociologists, philosophers and others dedicated to astrosociology standing by, ready to grapple with the meaning of the message and perhaps to reply in kind.

Third, those sensitive to the human condition could help temper the impact on our society of contact. Along with the SETI scientists who would explain to the public the nature of the signal and what it told us about the technology which produced it, we would need specialists in astrosociology to inform the public of the apparent nature of the sending civilization, and to involve them in the long process of learning about our distant neighbor and of adjusting to our newly realized status as but one of many natural experiments in intelligent life.

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